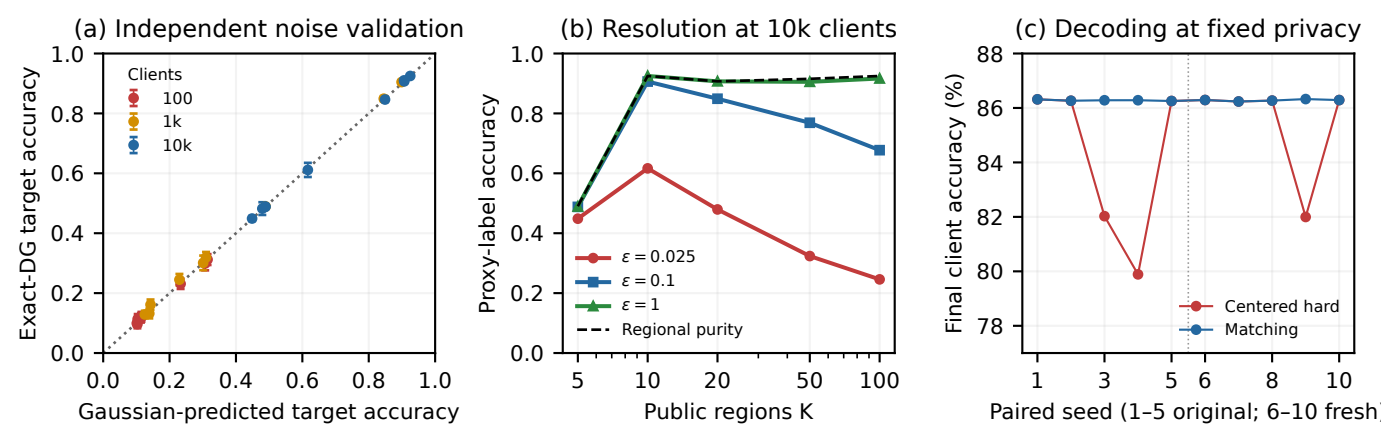
(a) Independent noise validation
Clients
100
1k
10k
Exact-DG target accuracy
Gaussian-predicted target accuracy
0.0
0.2
0.4
0.6
0.8
1.0
(b) Resolution at 10k clients
Proxy-label accuracy
Public regions K
5
10
20
50
100
ε = 0.025
ε = 0.1
ε = 1
Regional purity
(c) Decoding at fixed privacy
Final client accuracy (%)
78
80
82
84
86
88
1
3
5
6
8
10
Centered hard
Matching
Paired seed (1–5 original; 6–10 fresh)